

Saejoon Park

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EDUCATION

Cornell University Ithaca, New York

Aug 2024 - May 2028

Bachelor of Science in Electrical and Computer Engineering (Robotics Concentration)

SKILLS

Programming Languages: Python, C, C++, Swift

Software/Platforms: GitHub, KiCad, LTSpice, MuJoCo, Fusion360, Claude Code

Languages: English (fluent), Korean (fluent)

PROJECTS

Computer Vision + Glove-based Tracking Computer Mouse | Ithaca, NY

Jan 2025

- ATmega32U4; custom conductive inputs; IMU; assembled a position tracking glove device (Valorant compatible)
- Computer Vision (OpenCV) allows for hand gesture tracking(simulated clicks/volume control) at 30 fps

Thermal Electric Generator Powered Mouse | Fairfax, VA

Aug 2023

- Patent-pending hand thermal energy-powered computer mouse
- Implemented TEG Modules and a transformer to convert a 20mV output from TEG to 3.3V

Steel Electric Bike from Scratch | Burke, VA

May 2025 - Aug 2025

- Custom-cut and assembled steel chassis; 1000W motor; 50V accumulator

Audio Signal Decoding Circuit & Filter Design | Ithaca, NY

April 2026

- Designed and built a 6th-order MFB bandpass filter to extract encoded audio secrets.
- Conducted automated frequency response sweeps and eliminated multi-volt DC offsets safely.

Sensory Overload Wearable | Ithaca, NY

Feb 2025 - Now

- Developing a custom PCB for a lap wearable, accommodating individuals with sensory processing disorder
- [this belongs under Cornell Assistive Tech Project Team]

Political AI Camera | Ithaca, NY

Nov 2025 - Now

- Developed the Raspberry Pi-based hardware featuring a 12.1 MP sensor and software using the Picamera2 library
- Calls OpenAI API with the gpt-image-1 model, prompted to output different political perspectives of given image
- [this activity belongs under Cornell Machine Poetics Research]

Oddity 1 by IBX, Chrome Extension | Ithaca, NY

Feb 2026 - March 2026

- Using chenglou/pretext JS library, GDocs API, and HEX color tracking system, created a strategy for on-screen GDocs annotations (previously not possible w/o Google Whitelist), solving <canvas> rendering difficulties.
- Developed a “Cursor for Writing” system in GDocs using a clean-room rewritten Claude Code opensource code
- [this activity belongs under IBX LLC]

ACTIVITIES

Cornell Assistive Technologies Project Team | Ithaca, NY

Feb 2025 - Apr 2026

- Project-based engineering group for developing assistive technologies for people with disabilities

Cornell Machine Poetics Research, Sang Leigh | Ithaca, NY

Nov 2025 - Apr 2026

- Explores how expressive technology and AI can fit into products and systems to enhance human experience

WORK EXPERIENCE

Wim Robotics Internship

May 2026 - Aug 2026

- [Insert info here about Analog NN and UR5e control project]

Co-Founder/COO of IBX LLC

Mar 2025 - Now

- Currently Frontend Developer/Outreach: working with FCPS CTIO and Director of Counseling for app approval

HONORS

USPTO Provisional Patent for TEG Hot Swap Computer Mouse

Fall 2023